

Department of Chemical Engineering
Chemical Reaction Engineering II (CL24303)

MO2026

Tutorial 5

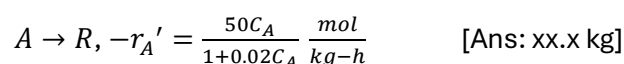
Date Assigned	29.09.2026	Submission Date and Time	15.10.2026 23:59 h
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- 1 It was wrongly assumed that for a certain catalytic reaction $A \rightarrow R$, the pore diffusion can be neglected, and thus the rate equation is:

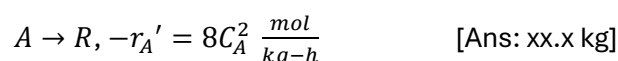
$$(-r_A')_{obs} = 0.5C_A \frac{mol}{kg-h}$$

How much of the excess catalyst was used due to pore diffusion, in a packed-bed reactor for 75% conversion of 50 m³/hr of pure gaseous A ($C_{A0} = 15 \text{ mol/m}^3$) if the intrinsic rate constant was 0.25 h⁻¹? Given: $\rho_b = 1.6 \text{ g/cc}$, $d_p = 2.5 \text{ mm}$, $D_e = 2 \times 10^{-6} \text{ m}^2/\text{s}$ [Ans: xx.x kg]

- 2 It is assumed that for a certain catalyst, the pore diffusion can be neglected. How much of that catalyst is needed in a packed-bed reactor for 80% conversion of 1000 m³/hr of pure gaseous A ($C_{A0} = 100 \text{ mol/m}^3$) if the stoichiometry and rates are given by:



- 3 Repeat problem 1 if the rate is now given by



4. Repeat problem 2 if the reaction is limited by diffusion within the pores. Assume 'A' to be weakly adsorbed. Given: $\rho_b = 1.3 \text{ g/cc}$, $d_p = 3.0 \text{ mm}$ [Ans: xx.x kg]

5. Ex 15-3 on Page 746 of Fogler. Refer to section 15.7 for background

6. Using the [EUROKIN](#) open spreadsheet as a reference, develop a tool for designing gas-solid fixed-bed reactors. It should be used for industrial and academic members and promote best practices in chemical reaction kinetics, including experimental design, interpretation of kinetic data, assessment of experimental conditions, evaluation of kinetic-data-processing software, and dissemination of state-of-the-art knowledge through reviews and user-friendly tools. Explain how the spreadsheet can be used to assess transport limitations and to identify experimental conditions under which reliable intrinsic kinetic data can be obtained.

You may use AI to develop the spreadsheet. (Deadline: Day of your viva)